MOHANGRAPHY

Photography Portfolio Website

Technical Documentation

NC Mohan · mohangraphy.com · March 2026

Table of Contents

1. System Overview

Mohangraphy is a custom-built photography portfolio website running at **mohangraphy.com**. The entire site is generated from a single Python script and hosted on GitHub Pages — there is no CMS or backend server. All interactive features (visit counting, likes, subscriptions, email notifications) are powered by Supabase and Resend via browser-side JavaScript.

1.1 Technology Stack

	Technology
Site Generation	Python 3 (Claude_mohangraphy.py)
Hosting	GitHub Pages (mohangraphy/mohangraphy.github.io)
Database	Supabase (PostgreSQL via REST API)
Email Service	Resend (DNS verified on mohangraphy.com)
Analytics	Google Analytics GA4 + Plausible.io
Fonts	Cormorant Garamond (titles) + Montserrat (body)
Domain	GoDaddy — mohangraphy.com
CI/CD	GitHub Actions (.github/workflows/notify.yml)

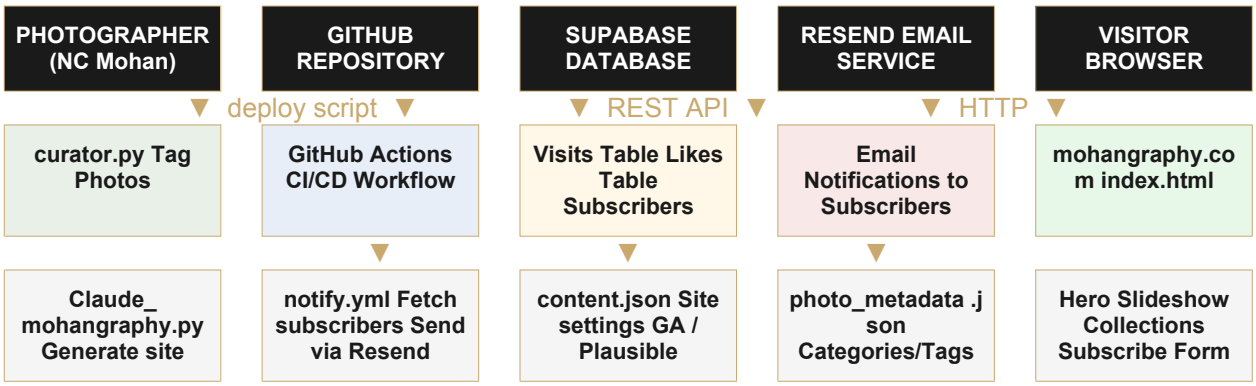
1.2 Key File Locations

	Purpose
Scripts/ Claude_mohangraphy.py	Main deploy script — generates full index.html
Scripts/curator.py	Photo tagging tool — runs on Mac before deploying
Scripts/content.json	Site settings, credentials, GA/Plausible IDs
Scripts/photo_metadata.json	All photo tags, categories, locations, dates
Photos/	Source photo files (.jpg/.png)
.github/workflows/notify.yml	GitHub Action — sends subscriber emails on push

2. System Architecture

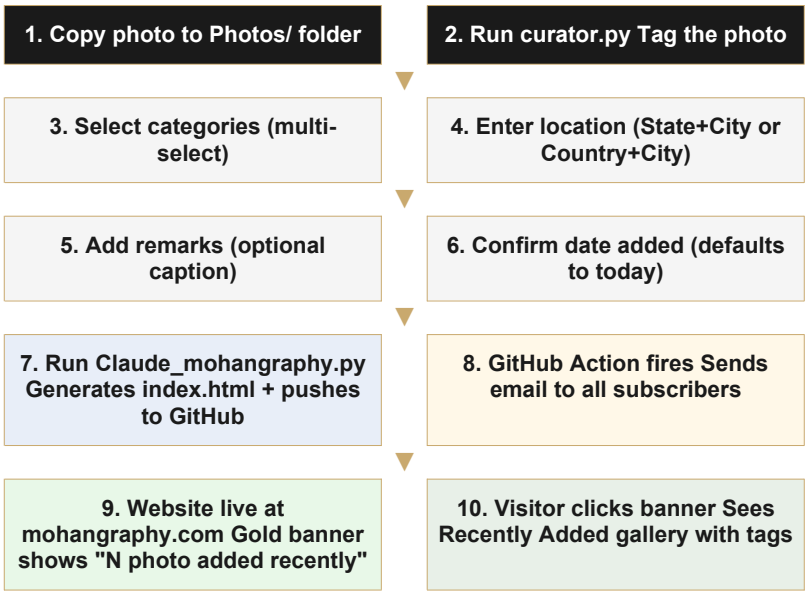
2.1 Overall Architecture

The diagram below shows how all components of the Mohangraphy system interact:



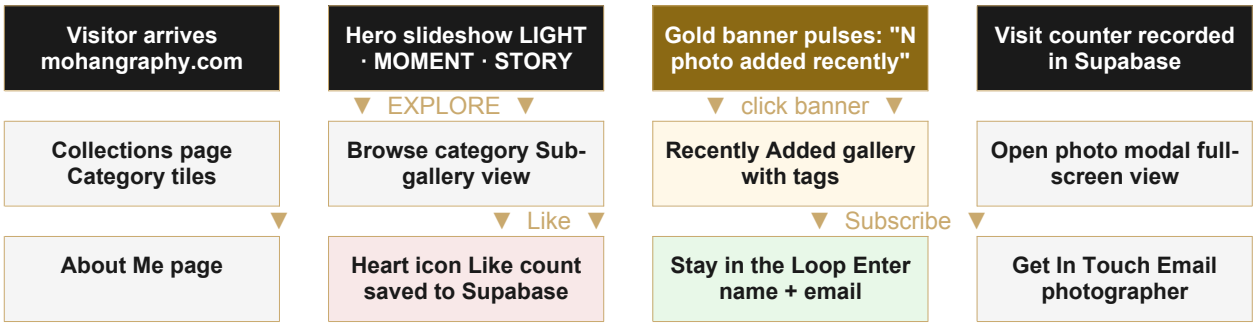
2.2 Photo Upload & Publish Workflow

This flowchart shows the complete journey from adding a new photo to it appearing live on the website with email notifications sent to subscribers:



2.3 Visitor Journey

This flowchart shows all the paths a visitor can take on mohangraphy.com:



3. Deploy Script (Claude_mohangraphy.py)

This is the heart of the system. Running this script regenerates the entire website from scratch and pushes it to GitHub. It is the only file you need to run when making any change to the site.

3.1 How to Run

```
cd /Users/ncm/Pictures/Mohangraphy && python3 Scripts/Claude_mohangraphy.py
```

3.2 What it Does

- Reads content.json for site settings and credentials
- Reads photo_metadata.json for all photo data
- Generates a complete index.html with all CSS and JS embedded
- Commits and pushes to GitHub (triggers GitHub Actions)
- GitHub Actions sends email notifications to all subscribers via Resend

3.3 Important Rules

- Never edit index.html directly — it is overwritten every deploy
- All permanent changes must be made inside Claude_mohangraphy.py
- Always run curator.py first when adding new photos

4. Photo Tagging (curator.py)

curator.py is a Mac GUI tool that opens each photo and prompts for metadata. It saves all data to photo_metadata.json which is then used by the deploy script.

4.1 How to Run

```
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/curator.py
```

4.2 Tagging Flow

- Each photo opens in Preview automatically
- Choose: Edit/Tag, Skip, or Stop
- Select one or more categories from the list
- Enter location — logic depends on category:
  - Places/National → asks for State then City
  - Places/International → asks for Country then City
  - Other categories → single Location field
- Enter optional remarks (caption shown as overlay on photo)
- Confirm date added (defaults to today's date)
- Data saved immediately to photo_metadata.json

4.3 Categories Available

	Description
Architecture	Buildings, structures, urban photography
Birds	Bird photography
Flowers	Floral and botanical photography
Nature/Landscape	General landscape photography
Nature/Landscape/ Mountains	Mountain and highland landscapes
Nature/Sunsets	Sunrise and sunset photography
Nature/Wildlife	Animals and wildlife (non-bird)
People/Portraits	Portrait photography
Places/International	International locations — asks Country + City
Places/National	Indian locations — asks State + City

4.4 photo_metadata.json Structure

```
{
  "abc123hash": {
    "path":       "Photos/subfolder/image.jpg",
    "filename":   "image.jpg",
    "categories": ["Places/International", "Nature/Landscape"],
    "state":      "Canada",          // Country for International
    "city":       "Banff",
    "remarks":    "Rocky Mountains in autumn",
    "date_added": "2026-03-03"
  }
}
```

5. Site Settings (content.json)

content.json controls all site-level configuration. Edit this file to change credentials, analytics IDs, or site text.

	Purpose
site.title	Browser tab title
site.plausible_domain	Plausible analytics domain (www.mohangraphy.com)
site.ga_measurement_id	Google Analytics GA4 measurement ID (G-XXXXXXX)
supabase.url	Supabase project REST API URL
supabase.anon_key	Supabase anonymous key for browser access
resend.api_key	Resend API key for sending emails (stored in GitHub Secrets)
photographer.name	Photographer name shown on site
photographer.email	Contact email for Get In Touch button

6. Visitor-Facing Features

6.1 Hero Section

- Full-screen slideshow of featured photos
- LIGHT · MOMENT · STORY tagline with photographer name
- EXPLORE button scrolls to Collections
- Gold pulsing banner appears when new photos added (last 14 days)
- Banner text: "N photo(s) added recently — view it/them"
- Clicking banner opens Recently Added gallery

6.2 Recently Added Gallery

- Shows all photos added within the last 14 days
- Deduplicates by photo path — a photo in multiple categories counts as 1
- Each photo shows gold tag labels indicating which collections it belongs to
- Example tags: PLACES / INTERNATIONAL · NATURE / LANDSCAPE
- NEW badge shown on thumbnails throughout site for 14 days

6.3 Collections

- Category tiles on home screen — one per photo category
- Clicking a tile shows all photos in that category
- Sub-navigation for nested categories (Nature → Landscape → Mountains)

6.4 Photo Modal

- Click any thumbnail to open full-screen modal
- Shows photo, title, location (state/country + city), remarks
- Like button (heart icon) — gold when liked, white/grey by default
- Like count stored in Supabase likes table
- Print size request button — opens email to photographer

6.5 Stay in the Loop (Subscribe)

- Elegant subscribe form at bottom of page
- Fields: Name (optional) and Email Address (required)
- Email validated before submission
- Subscriber saved to Supabase subscribers table
- Confirmation message shown on success

- Duplicate email returns friendly already-subscribed message

6.6 Visit Counter

- Counts every unique visit to the site
- Visit count displayed subtly on the page
- Owner mode prevents photographer's own visits from being counted
- Activate owner mode: visit mohangraphy.com?owner=yes on each device
- Must re-activate if browser data is cleared

7. Email Notifications

7.1 How it Works

- Triggered automatically when photos are pushed to GitHub (Photos/** path)
- GitHub Actions runs notify.yml workflow
- Workflow fetches subscriber list from Supabase
- Sends individual email to each subscriber via Resend API
- Email sent from: notifications@mohangraphy.com

7.2 DNS Configuration (Resend)

All three Resend DNS records must be verified on GoDaddy for emails to send:

	Status
DKIM TXT (resend._domainkey)	Verified ✓
MX record (send → feedback-smtp.amazonses.com)	Verified ✓
SPF TXT (send → v=spf1 include nses.com)	Verified ✓

7.3 GitHub Secrets Required

	Value
RESEND_API_KEY	API key from resend.com dashboard
SUPABASE_URL	Supabase project URL
SUPABASE_ANON_KEY	Supabase anonymous key

8. Analytics

8.1 Google Analytics GA4

- Measurement ID stored in content.json as site.ga_measurement_id
- Automatically embedded in index.html on every deploy
- IP anonymisation enabled by default
- To opt out (owner device): open browser console and run:

```
localStorage.setItem('ga_optout', '1')
```

8.2 Plausible.io

- Privacy-friendly analytics — no cookies, no GDPR issues
- Domain set in content.json as site.plausible_domain
- View dashboard at plausible.io

8.3 Supabase Visit Counter

- Custom visit counter stored in Supabase visits table
- Counts displayed live on the website
- Owner mode URL: mohangraphy.com?owner=yes (activate on each device)

9. Image Protection

Seven layers of protection prevent unauthorised use of photographs:

	How it Works
Right-click disabled	Context menu suppressed on all images
Drag prevented	ondragstart returns false on all images
iOS long-press blocked	Touch events intercepted on mobile
Text selection disabled	CSS user-select: none across the site
Keyboard shortcuts blocked	Ctrl+S, Ctrl+U, F12 intercepted
Watermark overlay	Invisible overlay div on each photo
Copyright banner	Visible copyright footer on all pages

10. SEO & Metadata

- Open Graph meta tags for social sharing (og:title, og:image, og:description)
- Twitter Card meta tags
- theme-color meta tag for browser chrome on mobile
- Structured description: "Photography portfolio by NC Mohan"
- Canonical URL set to https://www.mohangraphy.com

11. Troubleshooting

	Solution
Email not received	Check Resend DNS records — all 3 must be Verified. Re-run deploy to trigger fresh send.
Banner shows wrong count	Photo appears in multiple categories. Code deduplicates by file path — run fresh deploy.
Subscribe form does nothing	Check browser console for errors. Ensure Supabase URL and key are correct in content.json.
Recently Added page blank	Check photo_metadata.json has date_added field for the photo. Re-tag with curator.py.
NEW badge not showing	Verify date_added is within 14 days. Check photo was re-deployed after tagging.
Visit count increasing for owner	Run mohangraphy.com?owner=yes on each device. Check localStorage.getItem("mohan_owner").
Photo appears in wrong category	Re-run curator.py, select correct categories, redeploy.
Deploy script fails	Check Git credentials. Ensure you are in the correct directory. Run git status to check.

12. Quick Reference Commands

12.1 Daily Workflow — Add New Photo

```
# 1. Copy photo to Photos folder
# 2. Tag it
python3 /Users/ncm/Pictures/Mohangraphy/Scripts/curator.py

# 3. Deploy
cd /Users/ncm/Pictures/Mohangraphy && python3 Scripts/Claude_mohangraphy.py
```

12.2 Owner Mode — Exclude Your Visits

Visit this URL once on each device (Mac Chrome, Mac Safari, iPhone, iPad):

```
https://mohangraphy.com?owner=yes
```

To disable Google Analytics tracking on your device (browser console):

```
localStorage.setItem('ga_optout', '1')
```

12.3 Verify Fixes are Deployed

```
grep 'seenPaths' /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
grep 'subscribeVisitor' /Users/ncm/Pictures/Mohangraphy/Scripts/Claude_mohangraphy.py
```